DAIRYGUARD

Milk Collection Center Quality Control and Cold-Chain Spoilage Prevention System

INSY 8311

Database Development with PL/SQL

Capstone Project — Phase 0

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QUALITY

CONTROL

COLD-CHAIN

PROTECTION

Background and Problem

Who is affected

Smallholder dairy farmers delivering milk daily to Milk Collection Centers (MCCs).

MCC managers and cooperatives managing bulk chillers and farmer payouts.

Processors and consumers relying on safe, uncontaminated milk.

Why it matters

An estimated 15–20% of post-harvest milk may be lost daily due to delayed cooling and weak quality enforcement.

A warm (>7°C) or water-adulterated batch can cross-contaminate a bulk chiller.

Paper ledgers can cause recording errors, delayed payouts and flat pricing.

Motivation

Protecting farmer livelihoods: milk is a daily cash crop for rural Rwandan families; preventing batch spoilage protects household and cooperative income.

Bridging the enforcement gap: paper records and basic spreadsheets record deliveries but do not prevent unsuitable milk from entering a cooling tank.

Applying Oracle PL/SQL to real-world agri-tech: triggers, packages and live analytics can help address cold-chain quality challenges in Rwanda’s dairy sector.

Objectives

Main objective

Design and implement a normalized (3NF) Oracle PL/SQL database system.

Automate milk quality validation, cold-chain tank capacity tracking and quality-graded farmer payouts.

Specific objectives

Normalize milk quality records and cooperative structures into a 3NF relational schema.

Use PL/SQL triggers to enforce temperature, pH and density thresholds.

Automate monthly payouts with Grade A bonuses using packages and cursors.

Connect a live Power BI dashboard to monitor spoilage and tank utilization.

Research: Existing Solutions and Gaps

Existing solutions

Manual paper logbooks for morning/evening milk weights and lactometer notes.

Basic SMS or spreadsheet weight trackers, including RDCP pilot examples, for basic bookkeeping.

Identified gaps

No automated rejection logic to block warm or adulterated milk from tank assignment.

No real-time tank-capacity enforcement, creating overflow risk during peak delivery hours.

No tamper-proof audit trail for edits to lab tests or payout records.

Data Source

Primary dataset: Milk Quality Prediction Dataset on Kaggle (cpluzshrijayan/milkquality), containing 1,059 laboratory observations.

Contextual data: NISR Rwanda livestock statistics to model cooperative volumes and RWF payout rates.

Privacy: the source describes the dataset as anonymized sensor data with no personally identifiable information.

pH	Temp (°C)	Taste	Odor	Fat	Turbidity	Color	Quality
6.6	35	1	0	1	0	254	High (Grade A)
6.6	36	0	1	0	1	253	Medium (Grade B)
8.5	70	1	1	1	1	246	Low (Rejected)
3.0	40	1	0	0	0	255	Low (Rejected)

Proposed Solution and Innovation

Farmer delivers raw milk; technician records volume and rapid test readings (temperature, pH, density and fat).

Oracle PL/SQL validates quality thresholds and tank capacity before the transaction is committed.

Accepted milk updates COOLING_TANKS; rejected milk is blocked and the incident is logged.

A monthly PL/SQL process calculates farmer payouts and Grade A bonuses.



Innovation: Microsoft Power BI dashboard connected to Oracle PDB via ODBC/ODAC.

Planned Database: Seven Tables in 3NF

DISTRICTS_COOPERATIVES (coop_id PK) → one-to-many with FARMERS and COOLING_TANKS.

FARMERS (farmer_id PK, coop_id FK) → one-to-many with MILK_DELIVERIES and FARMER_PAYOUTS.

COOLING_TANKS (tank_id PK, coop_id FK) → one-to-many with MILK_DELIVERIES.

MILK_DELIVERIES (delivery_id PK, farmer_id FK, tank_id FK) → one-to-one with QUALITY_TESTS.

QUALITY_TESTS (test_id PK, delivery_id FK): temperature, pH, density, fat and grade.

FARMER_PAYOUTS (payout_id PK, farmer_id FK): monthly volume and quality bonuses.

AUDIT_LOG (audit_id PK): blocked deliveries and payout/test modifications.

PL/SQL Approach: Specific Use Cases

Triggers and validation

trg_validate_milk_quality: blocks tank entry if temperature >7°C or pH is outside 6.5–6.8.

trg_check_tank_capacity: prevents assignment if current volume + new liters exceeds capacity.

trg_audit_payout_changes: records UPDATE/DELETE changes to FARMER_PAYOUTS.

Stored program units

fn_calculate_quality_grade: returns GRADE_A, GRADE_B or REJECTED.

proc_record_delivery: validates farmer, grades milk, assigns a tank and updates volume atomically.

proc_generate_monthly_payouts: cursor-based monthly earnings and Grade A bonuses.

pkg_dairy_operations: groups routines and raises custom exceptions.

Expected Results: Phase VII Demonstration

Live database enforcement: insert a high-temperature or abnormal-pH batch and demonstrate trigger rejection before tank contamination.

Tank capacity protection: demonstrate blocking a delivery when the tank reaches its maximum liter capacity.

Automated payouts and auditing: run proc_generate_monthly_payouts and verify AUDIT_LOG entries after an attempted payout edit.

Live Power BI dashboard: demonstrate charts refreshing from Oracle PDB as new deliveries are recorded.

Limitations and Future Improvements

Current limitations

Rapid lab measurements (pH, temperature and density) are entered manually by the MCC technician.

Initial deployment models cooperative-level tanks rather than individual farm refrigeration units.

Future improvements

IoT integration: connect digital pH probes and chiller temperature sensors to Oracle through ORDS REST APIs.

Automated SMS alerts: use UTL_MAIL or an SMS gateway to notify farmers of rejected batches or approved payouts.

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